

DigitalBase Product Overview (Confidential)

Product Overview

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What is DigitalBase

DigitalBase is an on-premises LLM chat, RAG, and business-automation platform. Use AI safely on internal data without sending anything outside your organization.

Highlights

Fully on-premises

- All data stays inside your environment
- Operable without an internet connection
- No data is sent to the cloud by default (cloud LLMs such as OpenAI / Anthropic / Gemini can be optionally enabled via explicit configuration)

One-command install

- Supports macOS / Linux / Windows
- Install completes with a single command (**no Node.js required** — API and frontend run as a single process)
- Docker / Docker Compose are also supported

Multi-LLM engine

- **A single binary switches between three LLM backends via `LLM_BACKEND` in `.env`:**
 - **Ollama:** macOS / Linux / Windows (CPU and GPU both supported)
 - **vLLM:** Linux (NVIDIA GPU, high throughput)
 - **Cloud LLM:** OpenAI / Anthropic / Gemini can be used alongside, optionally (enabled via `.env`)

Enterprise authentication

- Local authentication (ID/password, bcrypt hashing)
- **LDAP / Active Directory integration** (python-ldap3)
- **OIDC / Azure AD (Microsoft Entra ID) integration**
- LDAP group → tag auto-mapping (shared directory ACL)

Branding customization

- Custom logo (text / image), custom title, color theme selection (8 themes)
 - Show/hide control of sidebar menu items
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Key features

AI Chat

- Multiple LLM model switching, conversation history management, multi-user support
- Streaming responses, Tool Calling, Thinking/Reasoning mode supported

RAG (Retrieval-Augmented Generation)

- Upload internal documents and let AI answer from them
- Supported formats: PDF, Word (.docx), Excel (.xlsx), Text, Markdown, CSV, JSON, Image (PNG/JPG/GIF/BMP/TIFF)
- **HNSW** vector search via pgvector (cosine similarity)
- Embedding model is switchable (default: embeddinggemma); chunk size / overlap configurable per Bot
- Create as a Bot and share inside the team with tag-based ACL + **runAs** (delegation of executing-user privileges)
- **Web search RAG** (DuckDuckGo / SearXNG, optionally enabled)
- RAG export / import (knowledge can be transported per Bot)

Document Creator

- Text extraction from PDFs and images (Vision LLM supported)
- Output as table / Markdown / JSON
- Excel / CSV import supported
- **Automatic document generation from templates**
- Folder-level batch processing (PDF / Word / image / CSV / JSON, etc. extracted and formatted sequentially)

File Hub

- Browse, edit, and connect to LLM tools across all internal files from a single UI
- **Four folder spaces:**
- **My folder** (`myfile/`): Personal to each user, invisible to other users
- **Shared folder** (`shared/`): Tag-based ACL (assign a tag to a shared directory → members holding that tag can view, edit, and read via LLM tools)
- **External folder** (`external/<label>/`): Safely browse existing OS paths exposed by the administrator via `.env` (`EXTERNAL_DATA_DIRS_READ` / `EXTERNAL_DATA_DIRS_WRITE`)
- **Connection folders** (`conn_<id>/` , `oauth_<id>/`): Personal OAuth / credential-based connections to remote storage
- **Remote storage connections** (configurable per user, dynamic forms in the UI):
- **Password / key authentication:** SFTP, SMB / CIFS (AD domain supported), FTP / FTPS (Explicit / Implicit), AWS S3 / S3-compatible, Google Cloud Storage, Azure Storage, WebDAV, HTTP
- **OAuth:** Google Drive, OneDrive, Dropbox, Box, SharePoint
- **Operations:** Hierarchical directory browse, upload (drag & drop), download, create, rename, delete, move, preview, edit shared ACL
- **LLM chat tool integration:** AI automatically invokes `list_files` / `read_file` / `search_files` / `describe_file` / `write_file` to answer using file contents (binary docx / xlsx / pdf are auto-converted to text)
- **Pipeline integration:** Connection-folder paths (e.g. `conn_<id>/sub/path`) can be referenced directly from operators (a connection created once is reusable across all features)
- **Permission guards:** Files cannot be placed directly under the shared folder (ACL is per directory); ADMIN can view and re-tag every directory; SUPER can retro-tag untagged directories

Pipeline (business automation engine)

- Combine **no-code operators** to automate business workflows
- Main operators:
- **File I/O:** reference any File Hub connection (S3 / GCS / Azure Blob / SFTP / FTP · FTPS / SMB · CIFS / WebDAV / HTTP / Google Drive / OneDrive / Dropbox / Box / SharePoint) directly from "Get file" / "Save file" (connections are shared with File Hub — create once, reuse everywhere), plus Google Sheets, rsync
- **API integration:** kintone, Salesforce, freee, MoneyForward, Sansan, HubSpot, SmartHR, Backlog, Notion, Jira, GitHub, GitLab, Shopify, Stripe and more
- **Data platforms:** BigQuery, ClickHouse, Snowflake, Elasticsearch, PostgreSQL (DataLake), REST/HTTP, RSS

- **AI processing:** LLM transform, AI classification, image understanding with AI (Vision), RAG load / search, document comparison
- **Data operations:** filter / set / sort / split / limit / aggregate / merge / compare / cast / dedup / validate / rename keys
- **Flow control:** IF / Switch / Loop / Wait
- Scheduled execution / Webhook trigger / manual execution
- Execution history and log management, exponential backoff and retry

MCP server / external AI agent integration

- **Built-in Model Context Protocol (MCP) server**
- DigitalBase RAG, Pipeline, and SQL can be used directly from Claude Desktop / Cursor / any MCP-compatible client
- Exposed via the `/api/mcp` JSON-RPC endpoint
- ExtAPI → MCP bridge lets external MCP clients invoke business systems

Helpdesk (internal inquiry management)

- Ticket creation, assignment, status management
- RAG / Bot integration automates first-line responses
- Multi-room (per department/project), unread notifications, Webhook notifications

SQL agent + Dashboard

- **Natural-language queries with AI** against external databases (PostgreSQL / MySQL / MariaDB / MS SQL / Oracle / Amazon Redshift / SQLite)
- Automatic retrieval of table lists and schemas, data editing, change tracking
- Connection information is saved and shared
- **Dashboard (Canvas-style)** — save SQL results as charts, place freely with drag/resize, 5 chart types (bar / line / pie / pivot / scatter) + value display / reference lines / data labels
- SQL refinement via AI chat (refine dialogue history is preserved)

Approval flow

- Multi-stage approval process, Webhook notifications, file attachments

Transcription (optional)

- Convert audio files to text (Whisper, tiny-large selectable)
- GPU support (Metal / CUDA, **including CUDA builds for RTX 50 Blackwell**)
- Supported formats: WAV, MP3, M4A, MP4, WebM, OGG, FLAC, AAC

Vision / OCR

- **Image understanding via Vision LLM** — Qwen3.6 / Gemma 4 / DeepSeek-OCR / Granite Vision, etc.: a single LLM extracts image contents (object recognition, table extraction, drawing description, handwriting OCR are all handled by a single LLM)
- OCR fallback: Tesseract (Japanese + English)
- Supported formats: PNG, JPG, GIF, BMP, WebP

Benchmark

- Performance comparison and evaluation of LLM models (speed / quality)

Prompt library

- Save, manage, and share prompts

User management

Role	Description
ADMIN	System administrator. Access to all features. User management, license management
SUPER	Sharing administrator. Tag management and assigning tags to users
USER	Regular user. Use of basic features

- Tag-based access control governs the sharing scope of Bots / Pipelines / SQL connections
- Shared Bots / Pipelines can be configured with `shareType` (PRIVATE / TAG / PUBLIC) and `runAs` (caller / owner)
- LDAP group → tag auto-mapping supported
- Feature restriction via sidebar menu customization

Authentication methods

Method	Description
Local authentication	ID/password (passlib bcrypt, 12 rounds)

Method	Description
LDAP	Active Directory / OpenLDAP (python-ldap3). Auto-creates a user on first login; retains 30+ LDAP attributes
OIDC	Azure AD (Microsoft Entra ID) supported. Auto-creates a user on first sign-in

- Even in LDAP/OIDC environments, the administrator (admin@local) can sign in via local authentication
- License-based user count limits

System requirements

Ollama edition (recommended)

Item	macOS	Linux	Windows
OS	macOS 12+	Ubuntu 20.04+	Windows 10+
CPU	Apple Silicon / Intel	x86_64 / ARM64	x86_64
Memory	8GB or more (16GB recommended)	8GB or more (16GB recommended)	8GB or more (16GB recommended)
Storage	10GB or more	10GB or more	10GB or more
GPU	Apple Silicon (Metal)	NVIDIA (CUDA) optional	NVIDIA (CUDA) optional

vLLM edition (high performance)

Item	Requirement
OS	Linux (Ubuntu 20.04+)
GPU	NVIDIA GPU required (CUDA 12.x, RTX 50 Blackwell supported)
VRAM	8GB or more (depends on model size)
Memory	16GB or more
Storage	20GB or more
Recommended hardware	NVIDIA DGX Spark, RTX 5060 Ti or higher

Required dependencies

Dependency	Purpose
PostgreSQL 17 + pgvector	Database + vector search
Ollama or vLLM	LLM engine
FFmpeg	Transcription (optional)
Tesseract OCR	OCR processing

Architecture

DigitalBase serves the **API and frontend (SPA) from a single process on a single port — 8000**. FastAPI returns both REST APIs and the Vite-built static files.

(図はシステム構成図を参照)

All data stays inside your environment (except when cloud LLMs are explicitly used)

Technology stack

Layer	Technology
Frontend	Vite + React 19 (static files) + Zustand + localStorage
Backend	FastAPI + uvicorn + SQLAlchemy 2.0+
Authentication libraries	python-jose (JWT) / python-ldap3 (LDAP) / passlib bcrypt
Database	PostgreSQL 17 + pgvector (DB name / user / password: <code>digitalbase</code>)
LLM communication	Direct HTTP requests to OpenAI-compatible endpoints via httpx (OpenAI SDK not used)

Network

Setting	Default / Note
API_HOST	Default <code>0.0.0.0</code> (LAN access allowed). Set to <code>127.0.0.1</code> to restrict to the same server
API_PORT	8000 (shared by API and Web)

In the Vite Edition, the Next.js-era "Web :3000 + API :8000" topology has been retired. The only externally exposed port is **8000**.

Installation

macOS:

```
curl -fsSL https://pub-a2cab4360f1748cab5ae1c0f12cddc0a.r2.dev/vite-scripts/install-macos.sh | bash
```

Linux:

```
curl -fsSL https://pub-a2cab4360f1748cab5ae1c0f12cddc0a.r2.dev/vite-scripts/install-linux.sh | bash
```

Windows:

```
irm https://pub-a2cab4360f1748cab5ae1c0f12cddc0a.r2.dev/vite-scripts/install-windows.ps1 | iex
```

- Install location: `~/.local/db` (shared by Ollama / vLLM — a single binary switched via `LLM_BACKEND` in `.env`)
- Start / stop: `db start` / `db stop`
- Deployment via Docker Compose is also supported